

A New Insight to the Visually and Auditory Oddball Task EEG Data Analysis

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University of Medical Sciences



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- *The views and opinions expressed in this presentation are those of the author(s) and do not necessarily reflect the original publications cited and used in this methodological research presentation.*

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Part Two (Practical)

Dataset

HPCA Step-by-Step

Part one

Problem

**Recent Literatures
(Special Issues 2020)**

**Out Result with
Hybrid PCA**

Part Two

Practical

The Dataset and the Experiment

The Analysis Method

Walz, Goldman, Carapezza, Muraskin, Brown, Sajda,
Conroy, Cheung Papers (2013 to 2015)

Functional Data Analysis

Fast Simultaneous Training of
Generalized Linear Models
(FaSTGLZ)

Simultaneous EEG-fMRI Reveals Temporal Evolution
of Coupling between Supramodal Cortical Attention
Networks and the Brainstem

Fast Bootstrapping and
Permutation Testing for
Assessing Reproducibility and
Interpretability of
Multivariate fMRI Decoding
Models

Prestimulus EEG alpha oscillations modulate task-
related fMRI BOLD responses to auditory stimuli

Simultaneous EEG-fMRI reveals a temporal
cascade of task-related and default-mode
activations during a simple target detection task

Function-on-function regression with thousands of
predictive curves
By Xin Qi , Ruiyan Luo (2018)

Hybrid principal components analysis for region-
referenced longitudinal functional EEG data
By AARON SCHEFFLER, DONATELLO TELESICA, QIAN LI,
CATHERINE A. SUGAR, CHARLOTTE DISTEFANO, SHAFALI
JESTE, DAMLA ,SENTÜRK (2020)



Volume 57, Issue 7
 Special Issue:
 50+ Years of P300:
 Where are We
 Now?
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 John Polich

Issue Information

Issue Information

Editorial

50+ years of P300: Where are we now?

Theoretical
Views

Components in the P300: Don't forget the Novelty P3! ←

Temporal PCA

A differential role for human hippocampus in novelty and contextual processing: Implications for P300

Temporo-Spatial PCA

Proceeding stimulus sequence effects on the *oddball-P300* in young and healthy older adults ←

Empirical
Topics

P3 amplitude reductions are associated with shared variance between internalizing and externalizing psychopathology

Significance? ... Significance! Empirical, methodological, and theoretical connections between the late positive potential and P300 as neural responses to stimulus significance: An integrative review.

The P3b and P600(s): Positive contributions to language comprehension

Comprehensive
Reviews

30+ years of P300 brain-computer interfaces

A systematic of physical activity and cardiorespiratory fitness on P3b

P300 development from infancy to adolescence

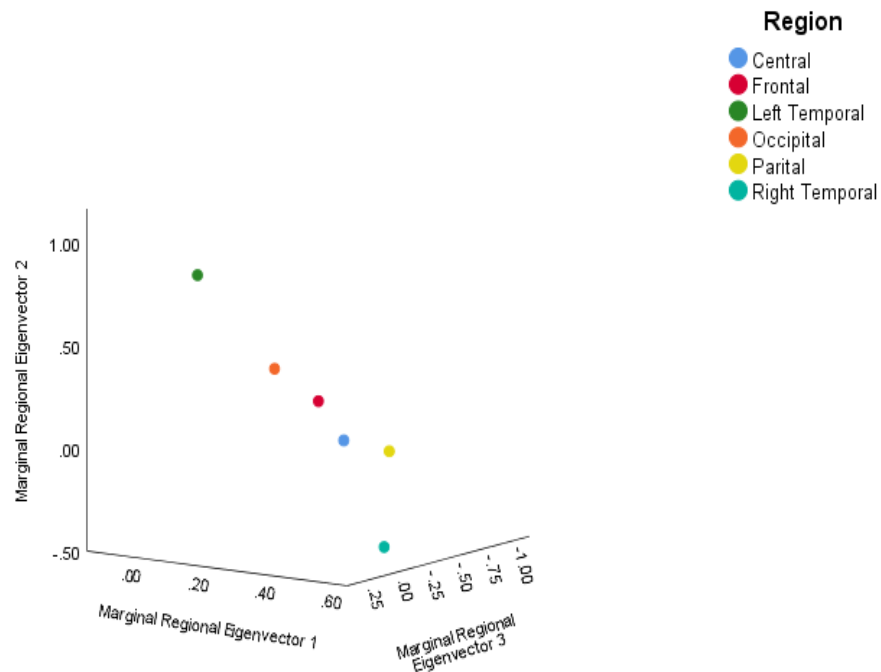
P300 in detecting concealed information and deception: A review

Effects of relevance and response frequency on P3b amplitudes:

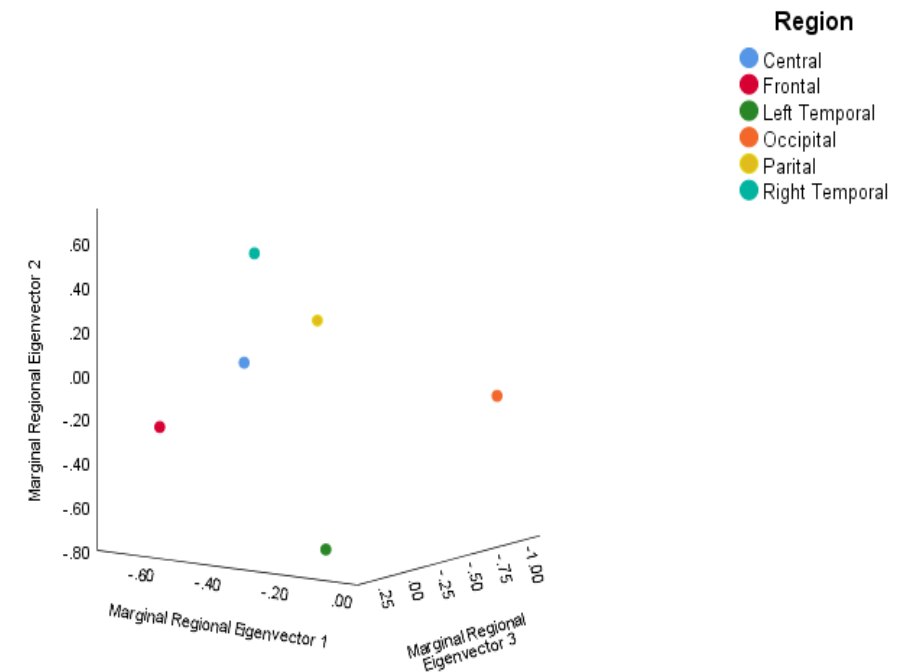
Review of findings and comparison of hypothesis about the process reflected by P3b

The Hybrid PCA

First Three Regional Marginal Eigenvectors



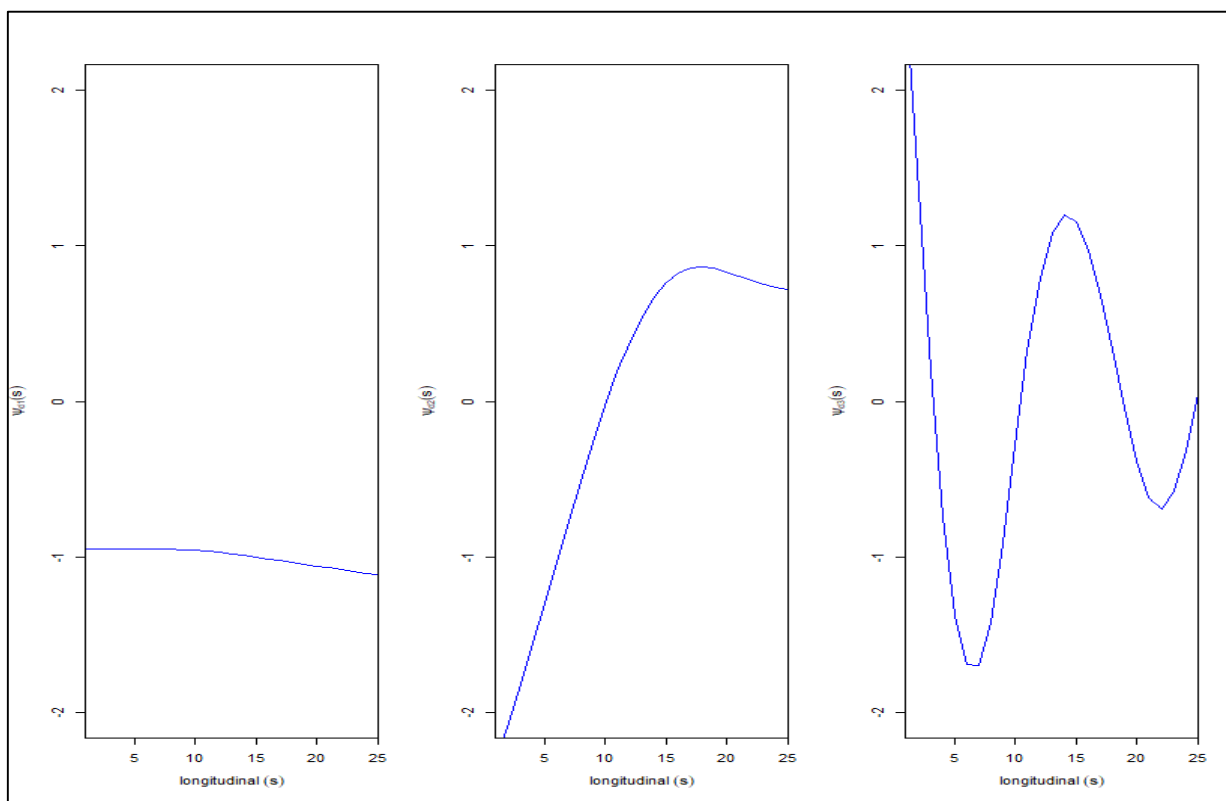
Target Auditory Stimuli



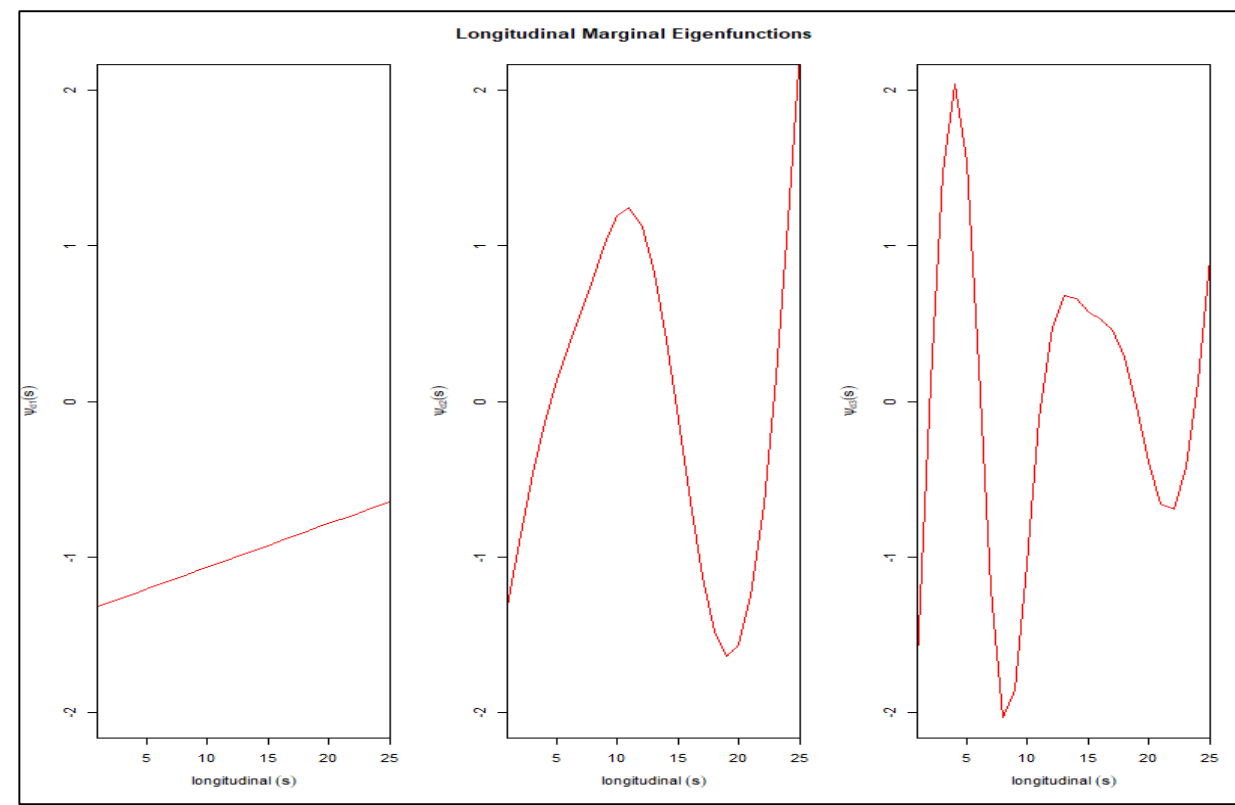
Target Visually Stimuli

The Hybrid PCA

First Three Longitudinal Marginal Eigenfunctions



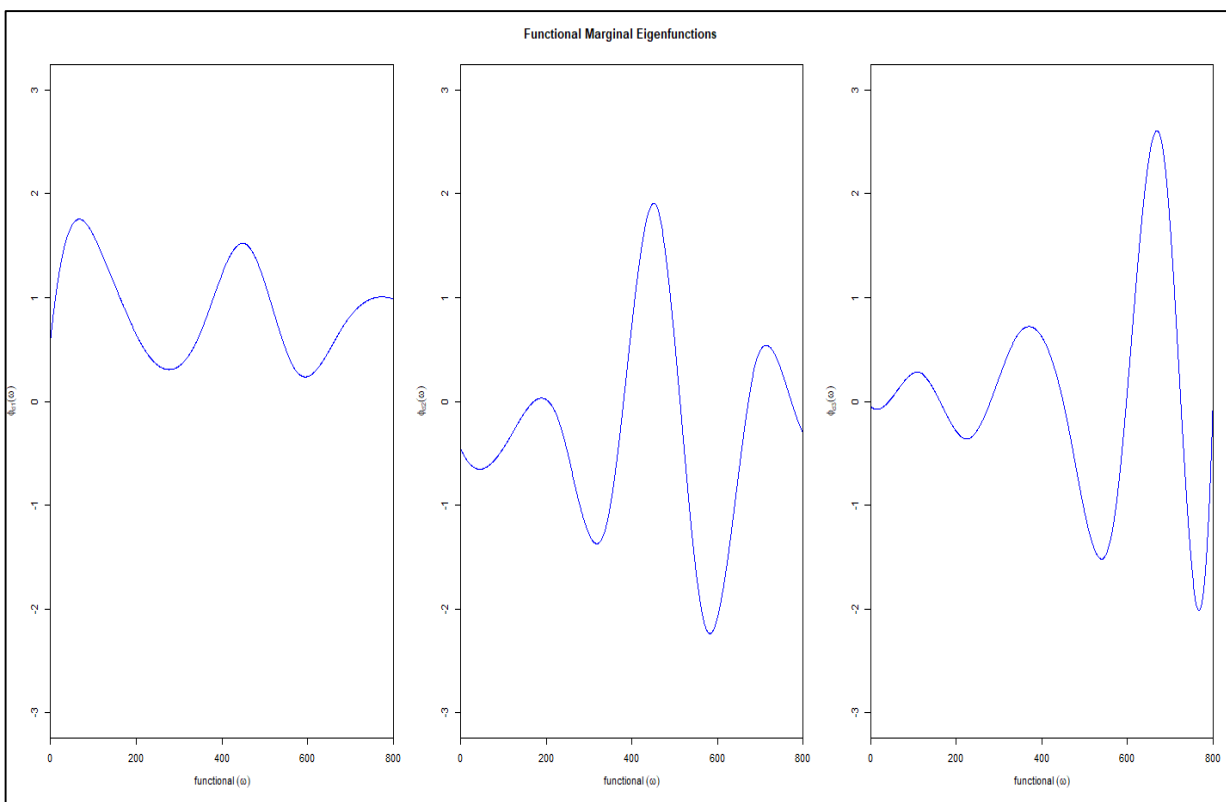
Target Auditory Stimuli



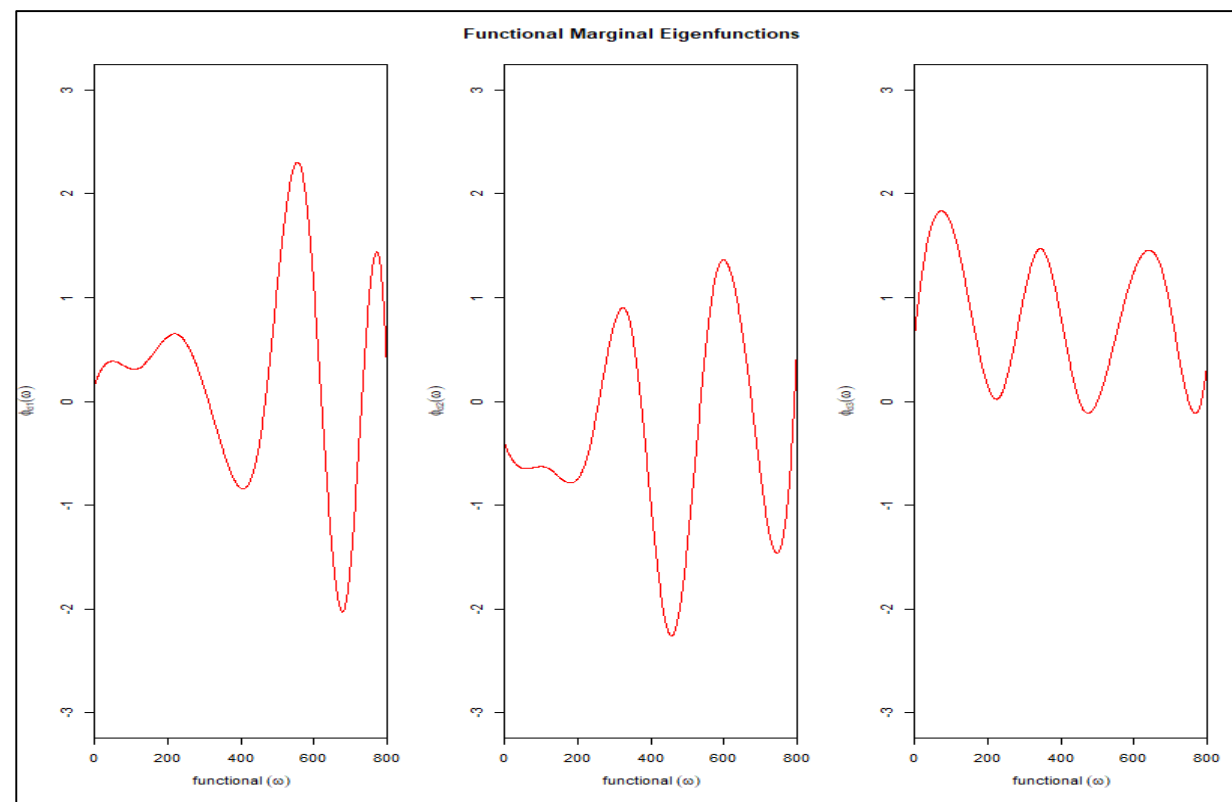
Target Visually Stimuli

The Hybrid PCA

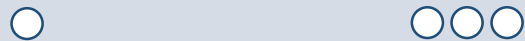
First Three Functional Marginal Eigenfunctions



Target Auditory Stimuli



Target Visually Stimuli

**Tasks:**

- **Three runs for each experiment**
- **Each run has 125 total stimuli**
 - **Auditory Stimuli**
 - **Standard Stimuli (390 Hz tone) – 80%**
 - **Target Stimuli (laser gun sound) – 20% - Press Button**
 - **Visually Stimuli**
 - **Standard Stimuli (small green circle on isoluminant grey background, 1.5 degree visual angle) – 80%**
 - **Target Stimuli (large red circle on isoluminant grey background, 3.45 degree visual angle) – 20% - Press Button**

Gender	N	Age mean(SD)	Dominant hand	Ethics
Male	11	26.7 (6.9)	Right: 8	Caucasian: 6 Hispanic : 2 Other: 3
Female	6	28.3 (4.7)	Right: 4	Caucasian: 5 Hispanic : 0 Other: 1

Table 1 – Demographic information of the participants

Total : 17 Healthy Subject**Sampling Rate :** 1000 Hz**Number of Channels :** 49

Custom Cap Configuration using bipolar electrode pairs

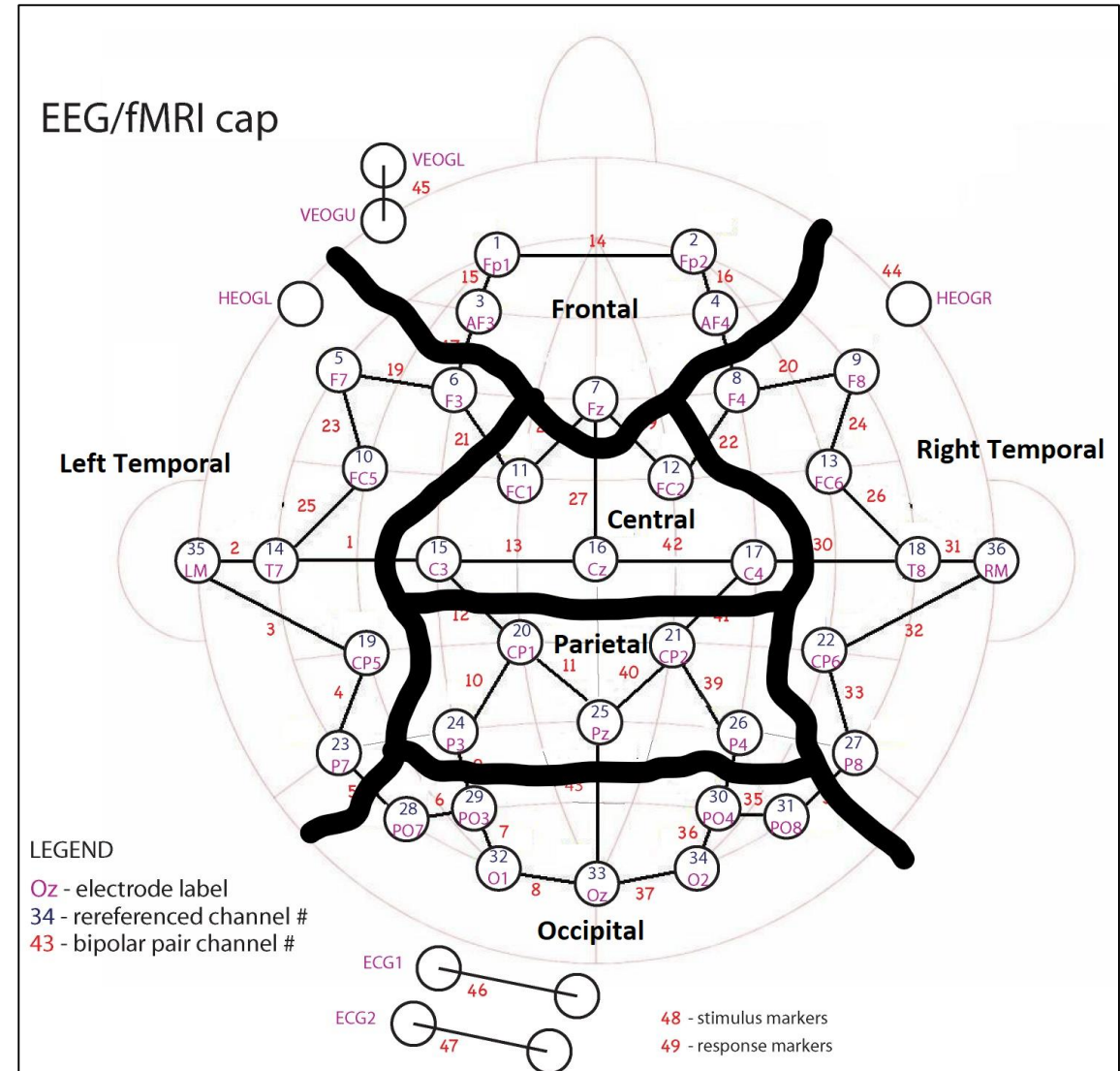
Analysis Channels: 34**Website:** <https://openneuro.org/>**Stimulus duration : 200 ms****ITI (inter-trial-interval) : 2-3 seconds****The first two stimuli is standard**

The six regions for averaging signals:

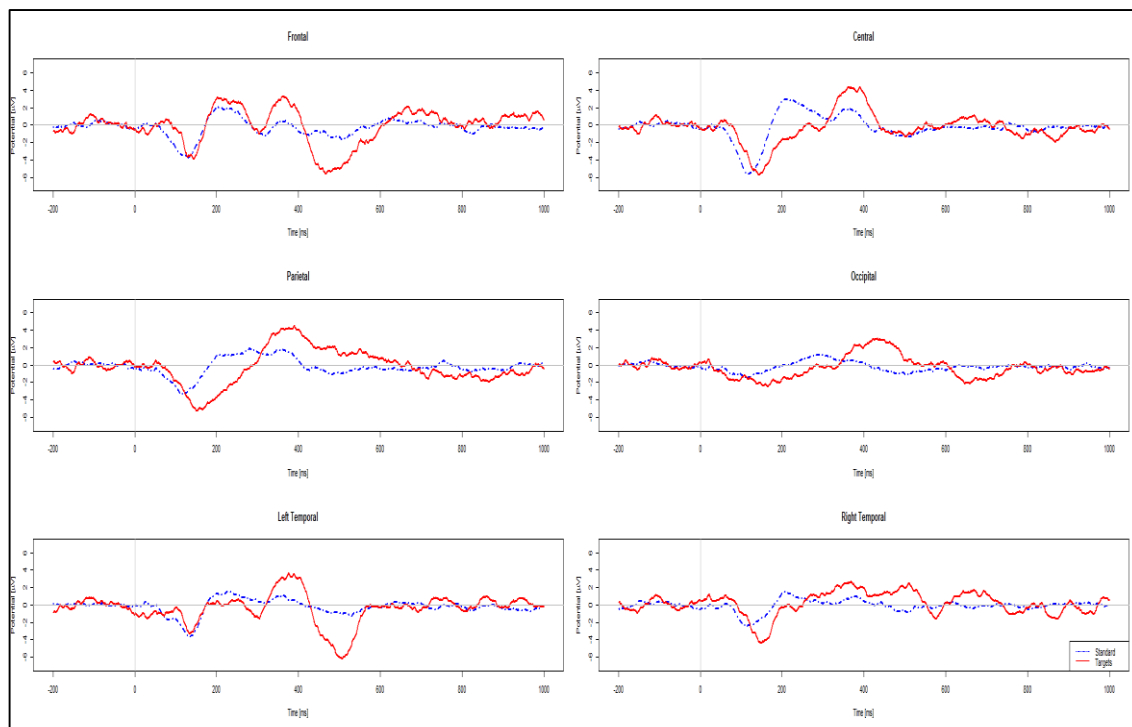
1. Frontal,
2. Central,
3. Occipital,
4. Parital,
5. Left Temporal and
6. Right Temporal.

There are two groups:

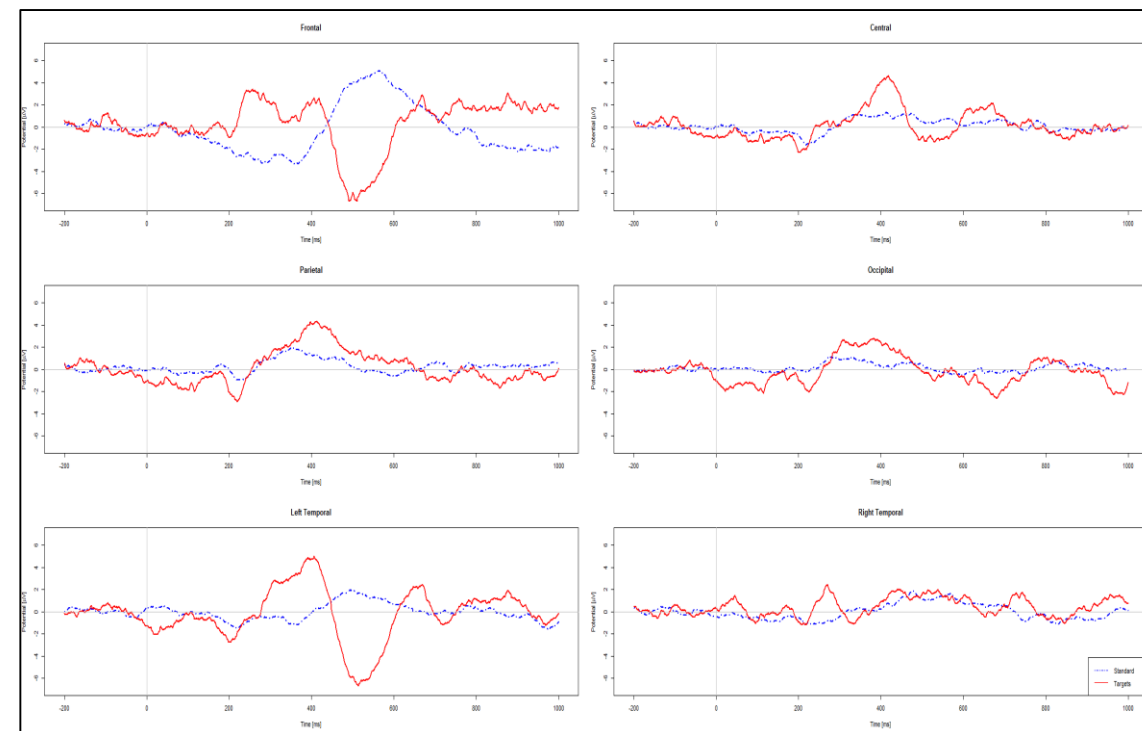
- Group 1: Target Stimuli (Auditory)
- Group 2: Target Stimuli (Visually)



Six Regions for Electrodes



The ERP (average over all single waves) by the brain regions for Standard vs Target Auditory stimuli



The ERP (average over all single waves) by the brain regions for Standard vs Target Visually stimuli

Qi, Xin, and Ruiyan Luo.

"Function-on-function regression with thousands of predictive curves."

Journal of Multivariate Analysis 163 (2018): 51-66.

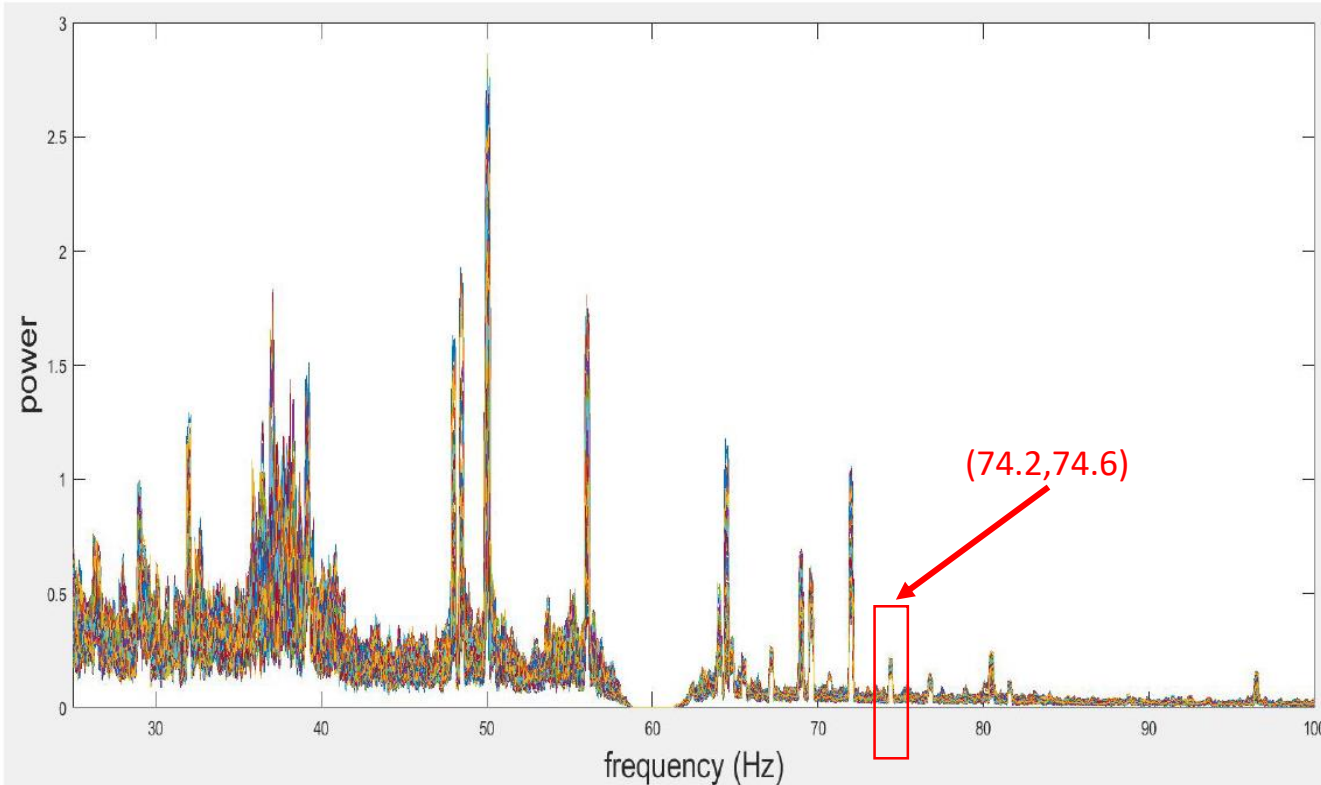


Figure S.15: The gamma spectra of the EEG data at 150 time points in one observation. Different color denotes the spectrum curves for different time points.

Response is Curve

The gamma frequency band (between 25 Hz and 100 Hz) is thought to be related to visual, auditory, and other sensory processing.

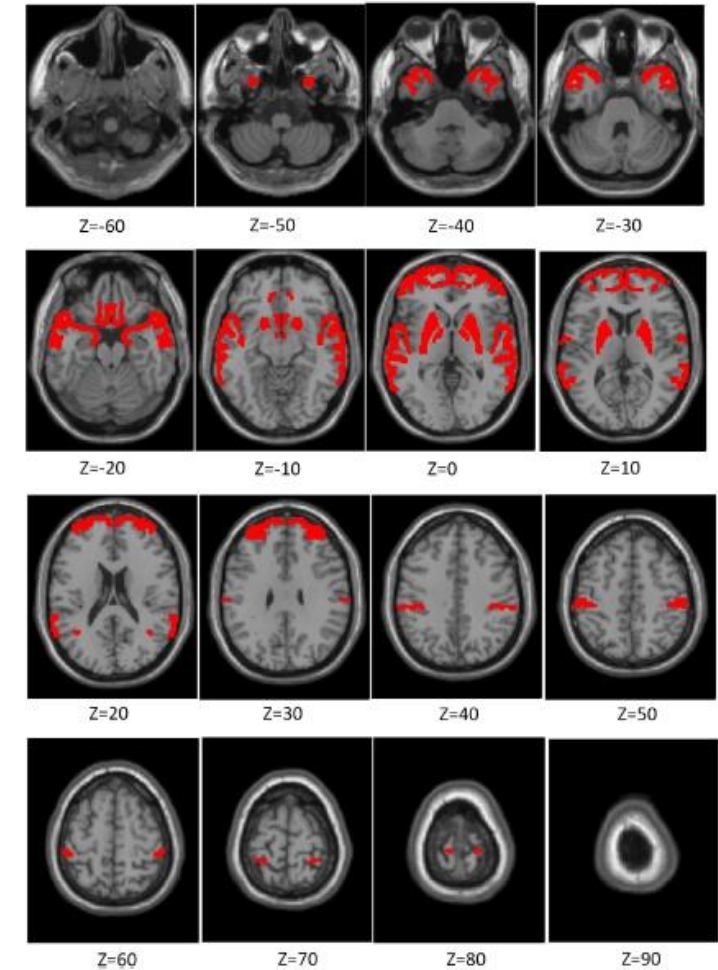


Fig. 4. The axial slices (which separate the bottom from the top of the brain) with the selected 11 ROIs highlighted for peak around 74.

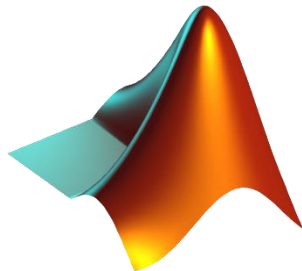
Covariates are ROI Curves



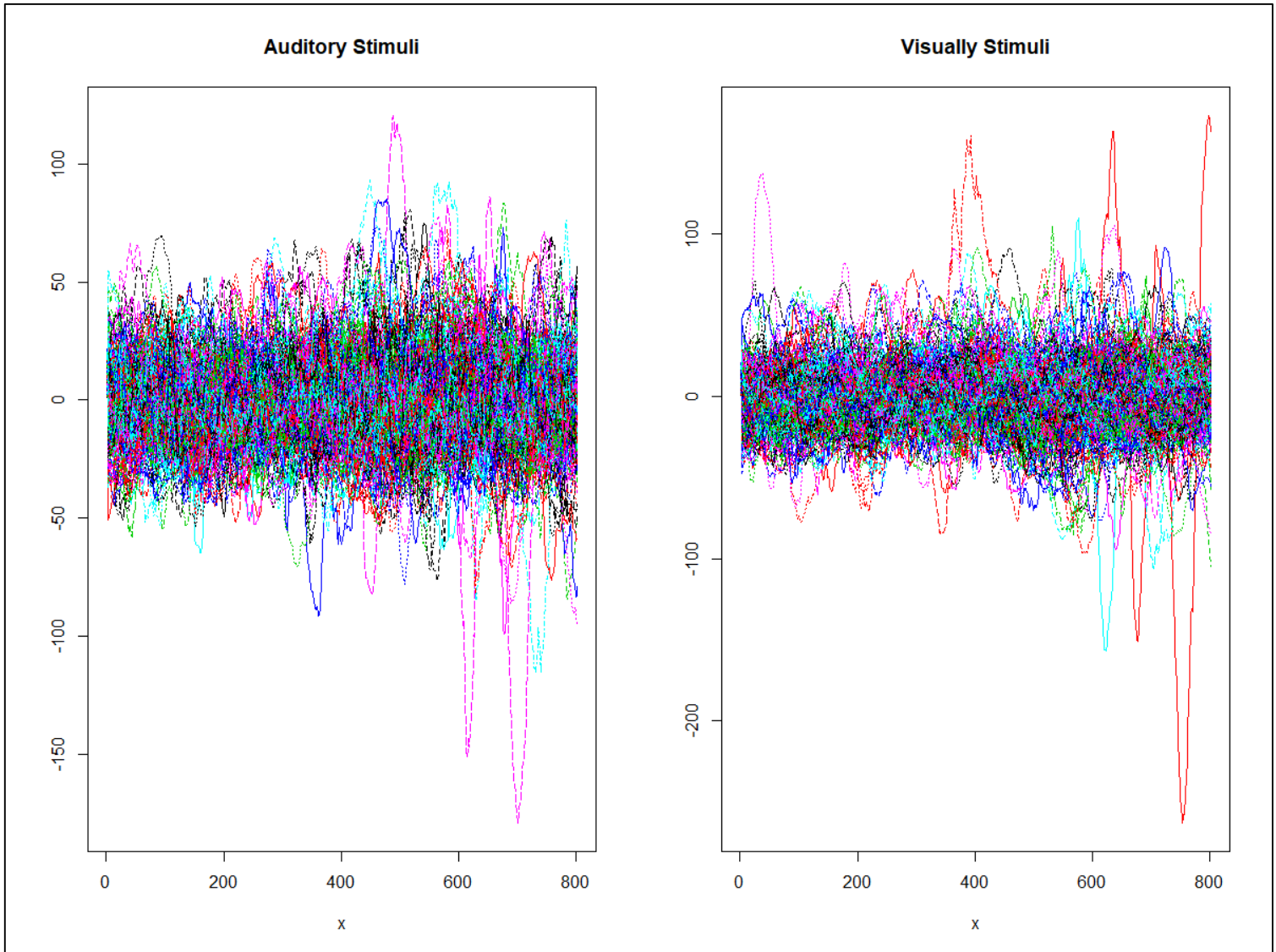
$Y_{di}(r, t, s)$ is ERP, which contains regional-longitudinal-functional data

- for subject i , $i = 1, \dots, 17$,
 - from group d , $d = 1$ (*Auditory Target*), 2 (*Visually Target*),
 - in region r , $r = 1, \dots, 6$,
 - at time t , $t \in [0, 800]ms$ for the functional part
 - s , $s \in [1, 25]$ for the longitudinal part.
-
- The hybrid principal component analysis was used to estimate the regional, longitudinal and functional effects.

- The dataset is cleaned
- The artifacts are removed
- The filtering is done
- The Epoching
- The baseline correction.



- All the codes are written in R version 3.6.3
- All the results are validated by EEGLAB Toolbox outputs.



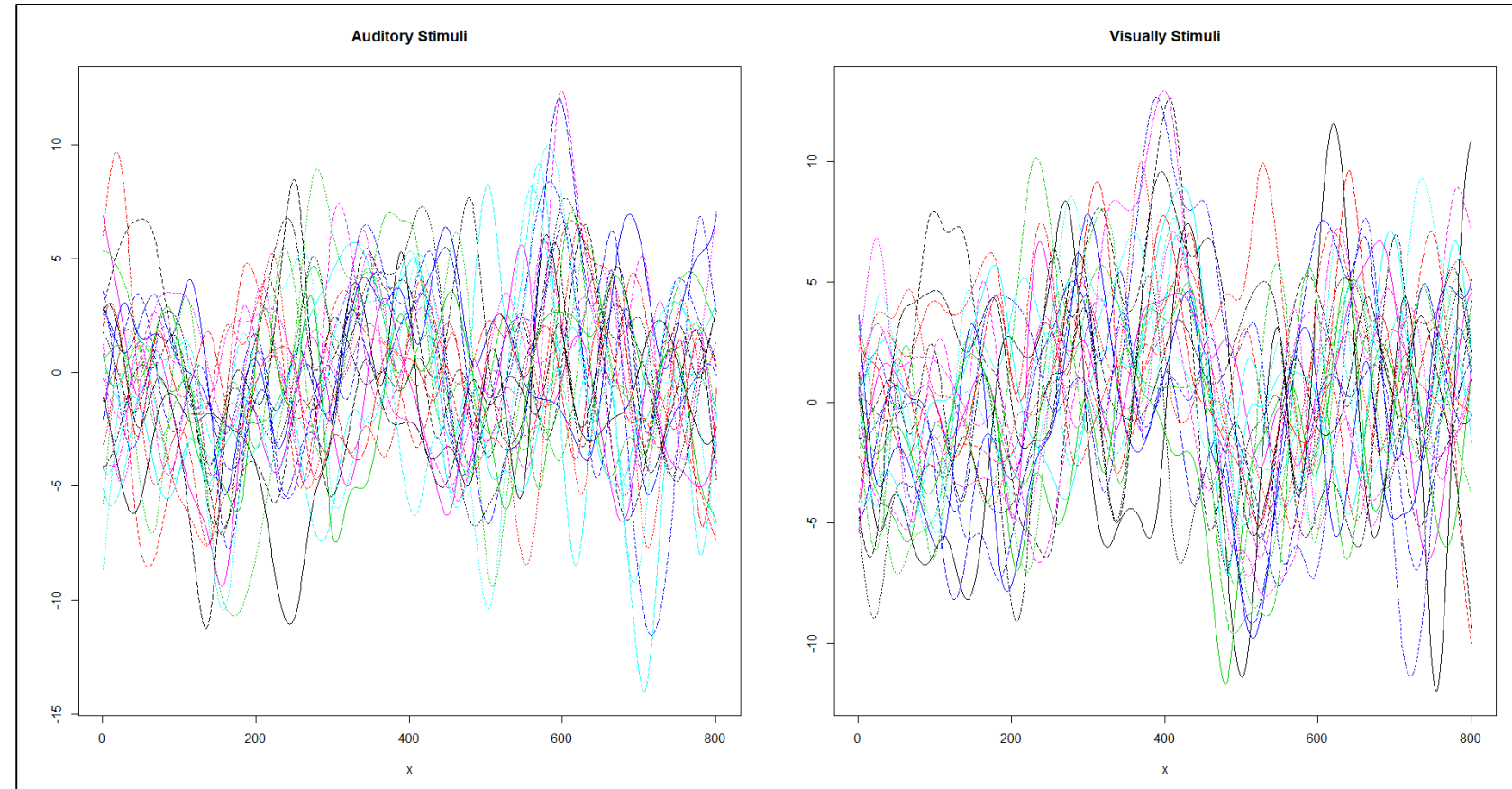
Uncleaned Epoched Data with some noises

Fast Bivariate P-Spline:

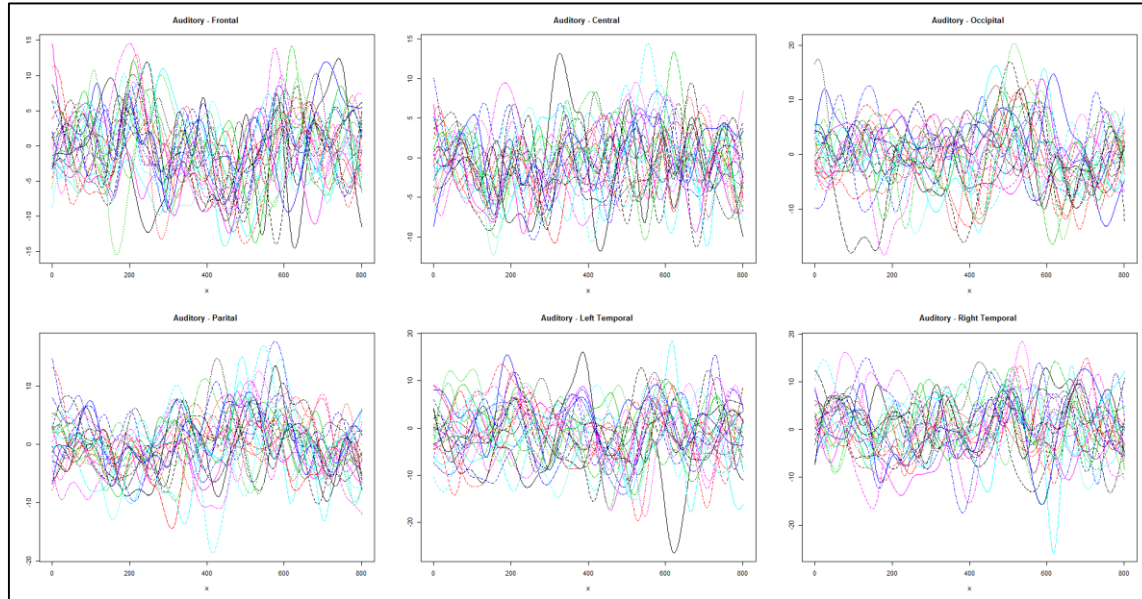
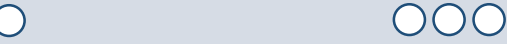
- Sandwich Smoother
- Fast
- Bivariate
- Selection of Smoothing Parameter by GCV (Generalized Cross-Validation)
- Optimization method: L-BFGS-B

Smoothing Two Domains:

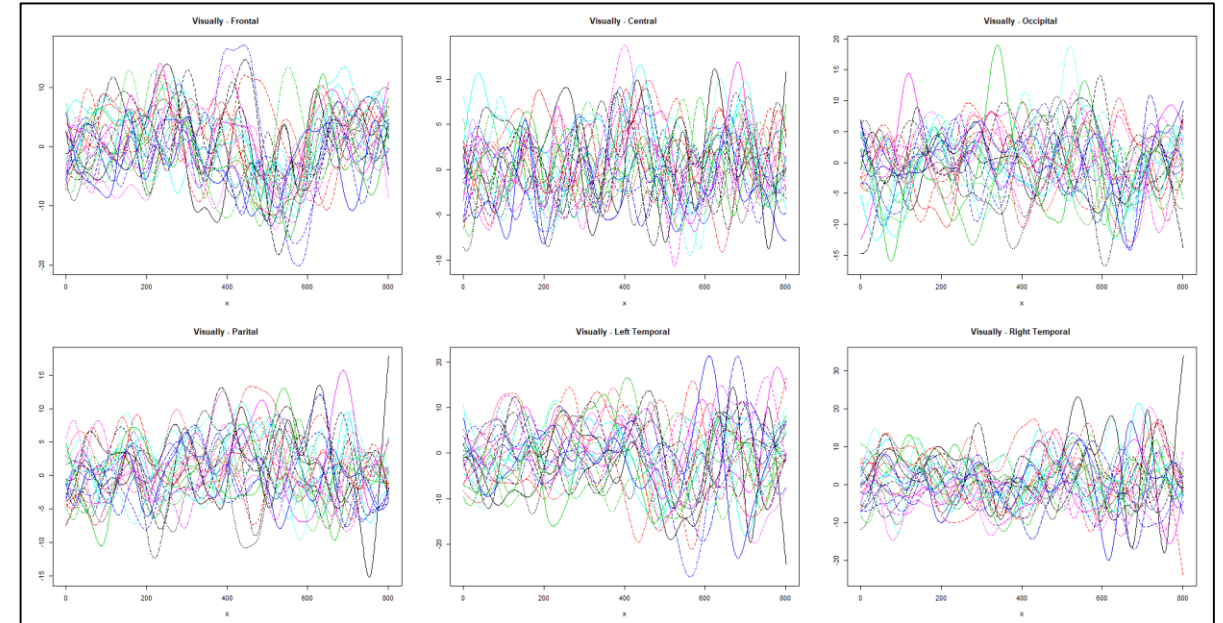
- Longitudinal Domain
- Functional Domain



Overall Mean Function for each Longitudinal Part $s \in [1, 25]$.



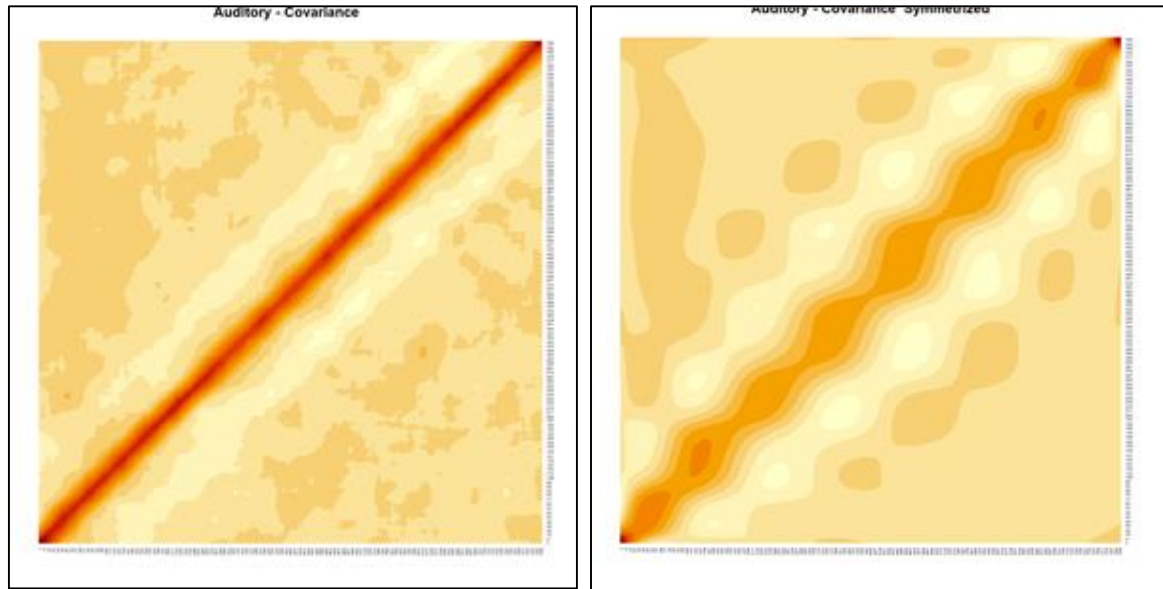
Estimated Group Region Shift for Auditory Stimuli
region $r, r = 1, \dots, 6$



Estimated Group Region Shift for Visually Stimuli
region $r, r = 1, \dots, 6$

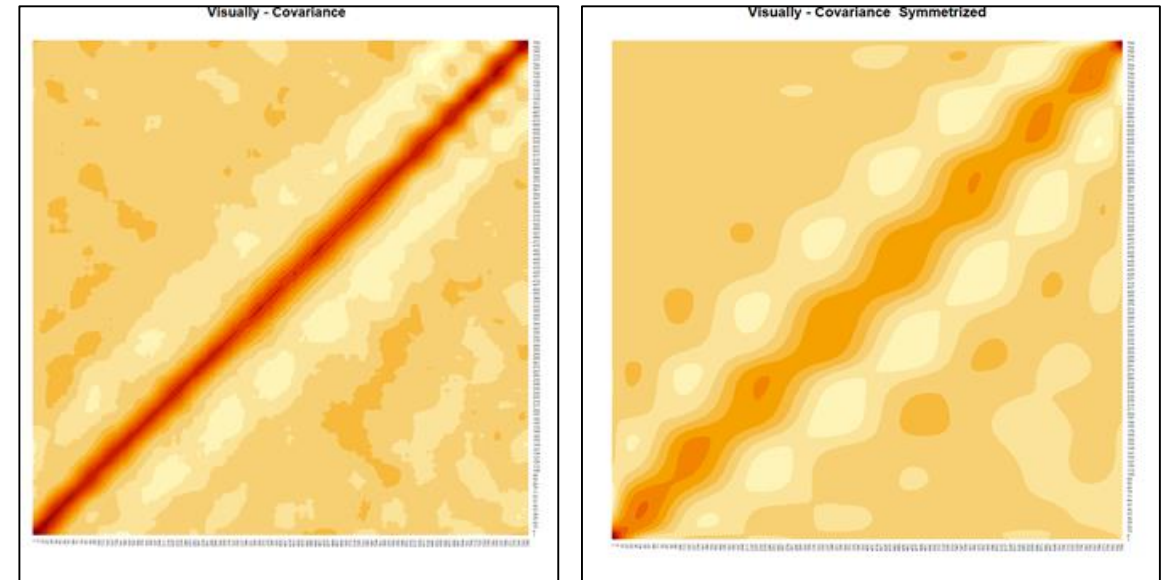
Estimated **Functional** Marginal Covariance

Group Auditory

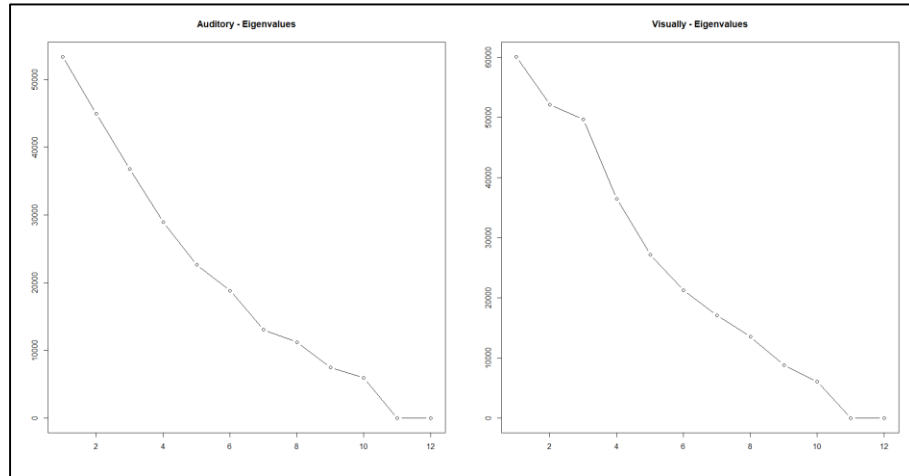
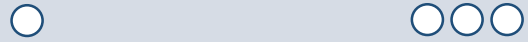


Measurement Error Variance: 250.7478

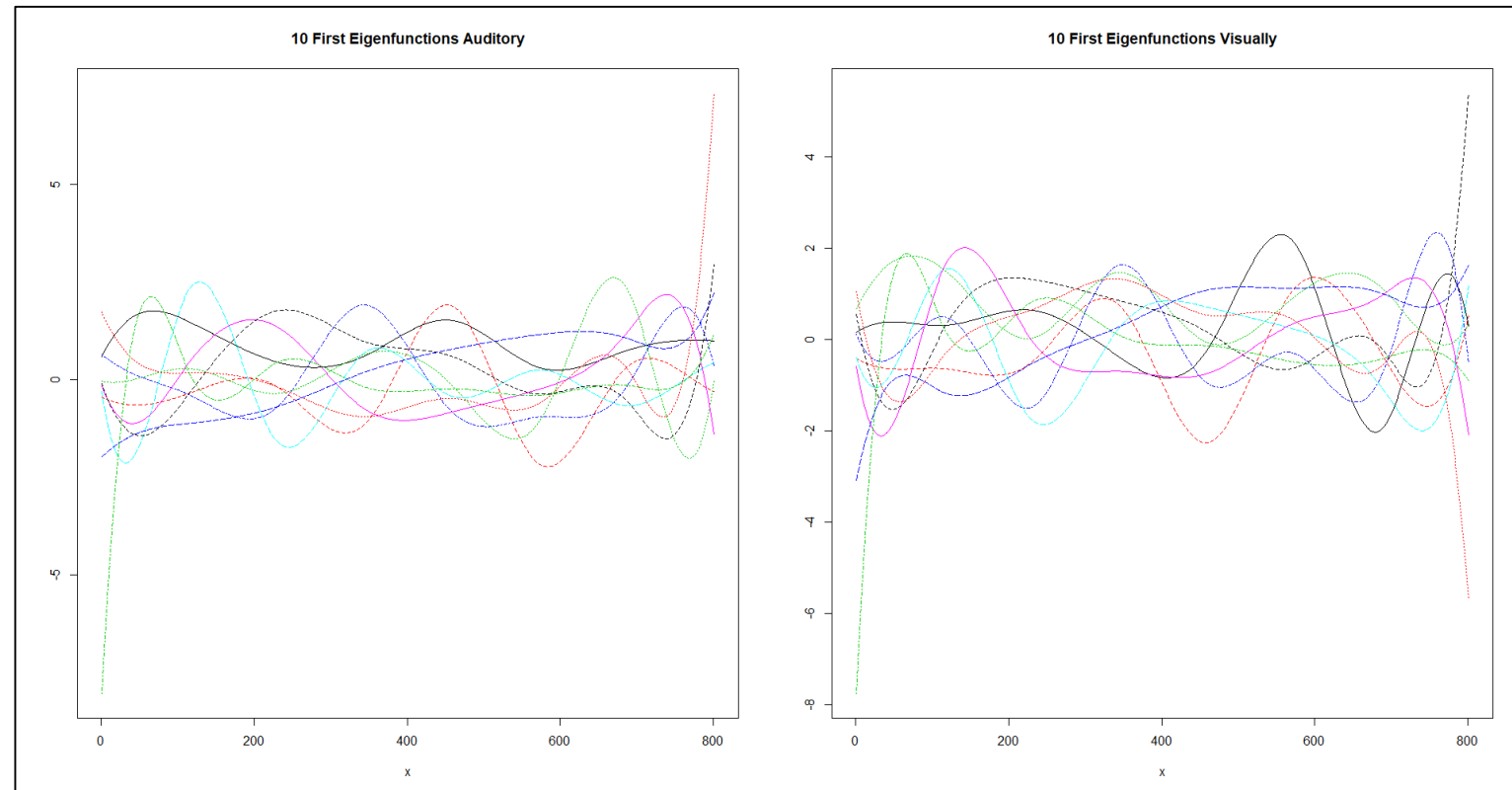
Group Visually



Measurement Error Variance: 272.0517



Number of Eigen Functions

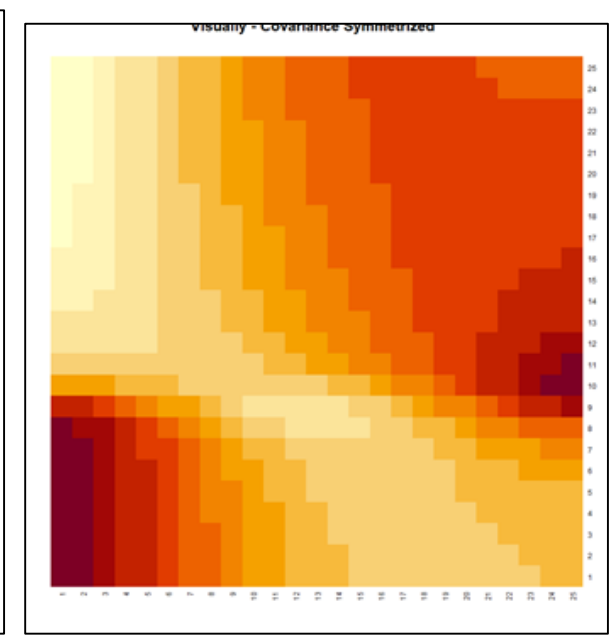
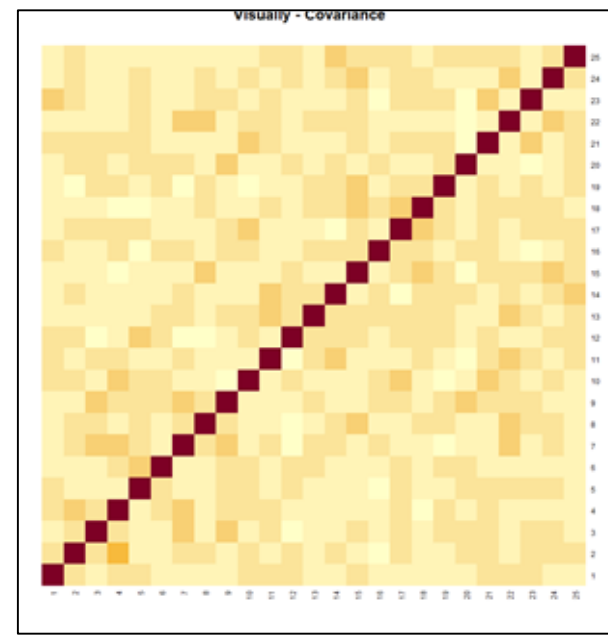
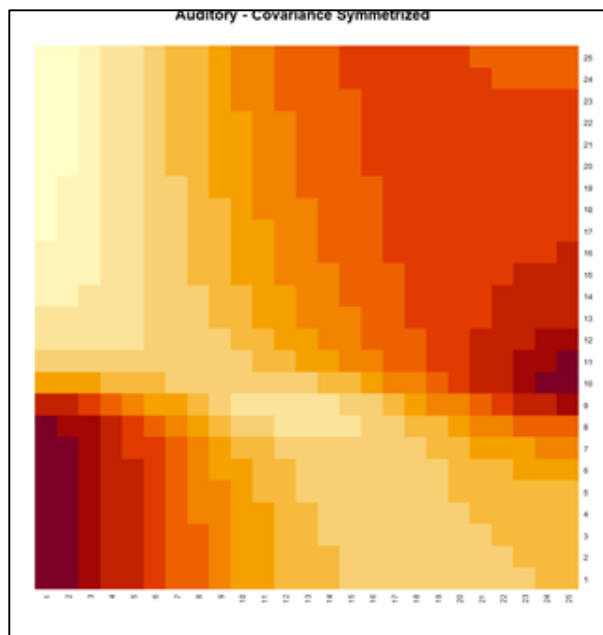
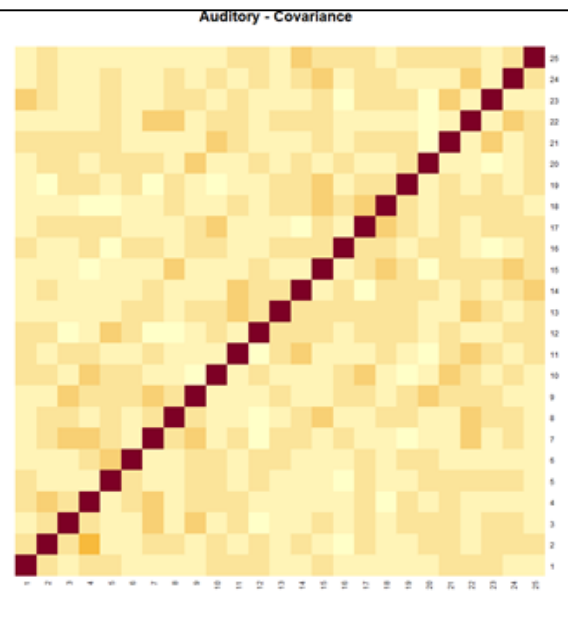
Estimated **Functional** Eigen Functions



Estimated **Longitudinal** Marginal Covariance

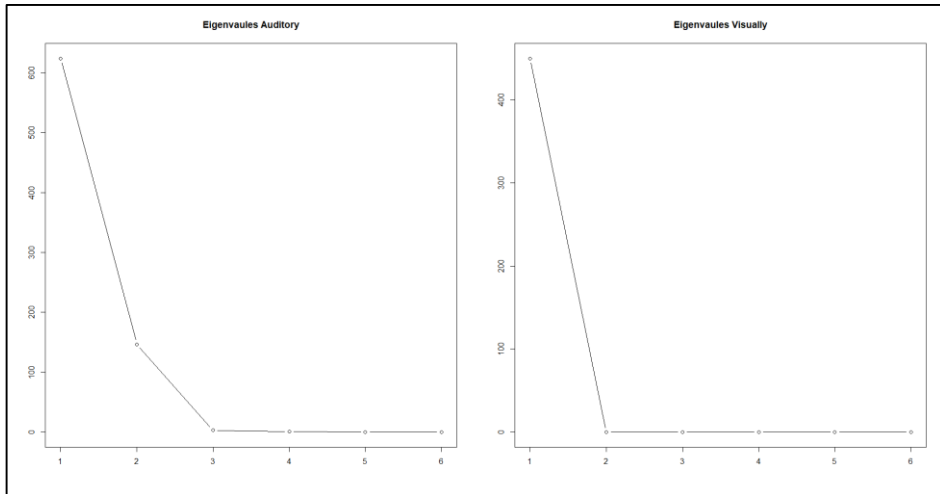
Group Auditory

Group Visually

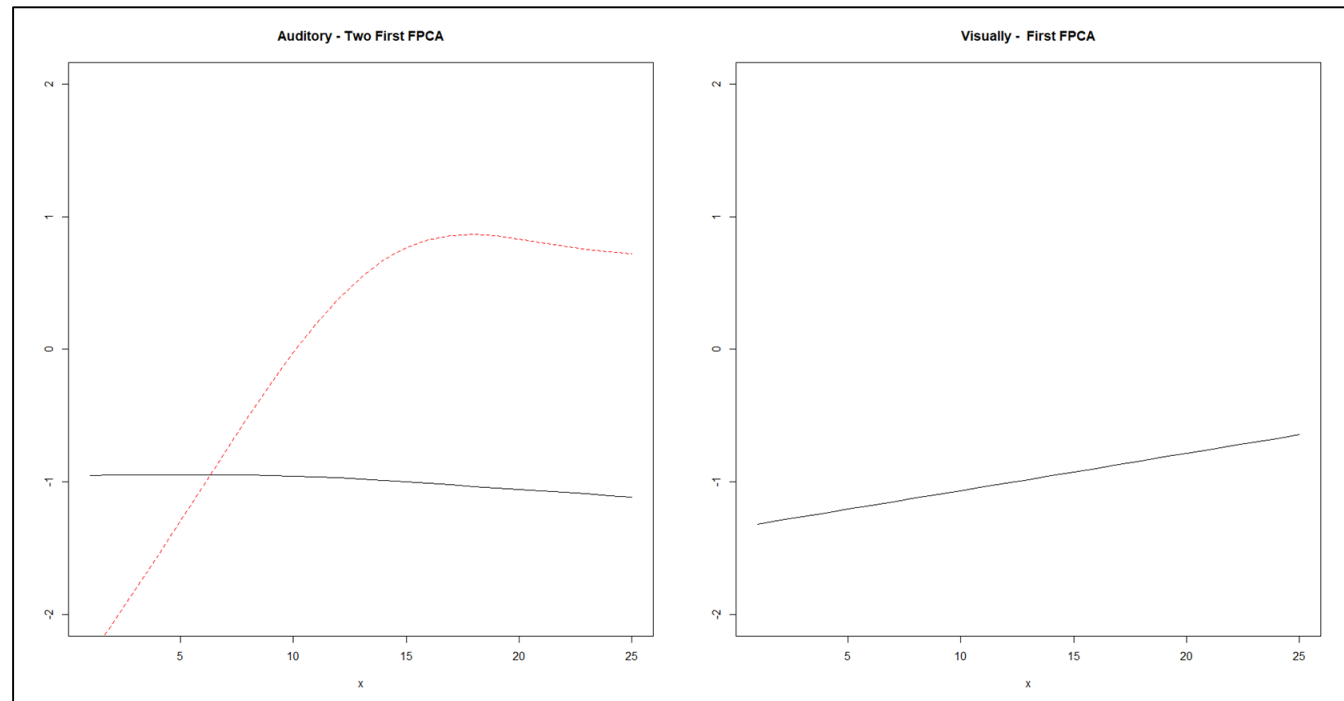


Measurement Error Variance: 527.2331

Measurement Error Variance: 622.4091



Number of Eigen Functions

Estimated **Longitudinal** Eigen Functions



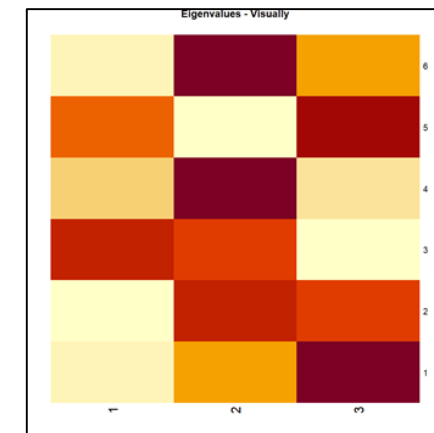
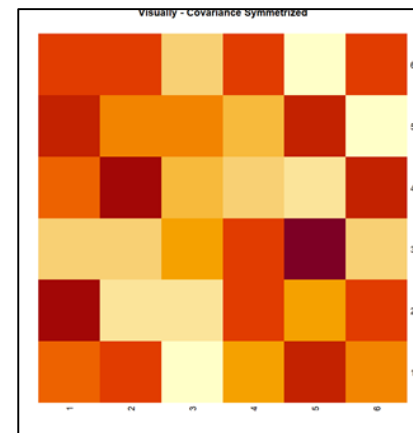
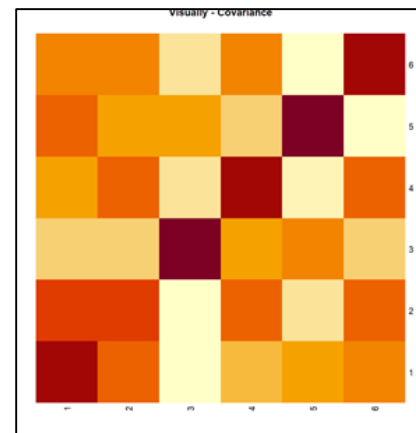
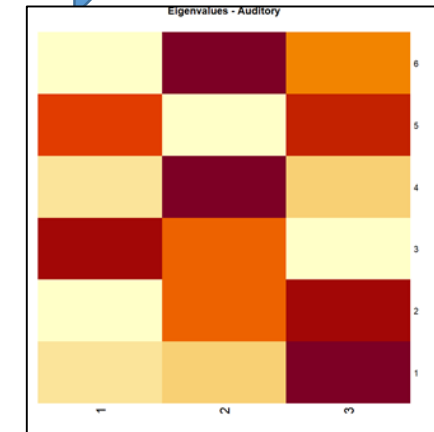
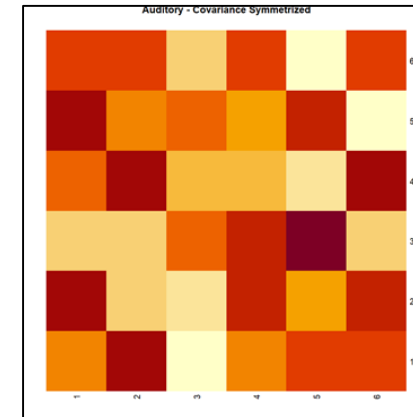
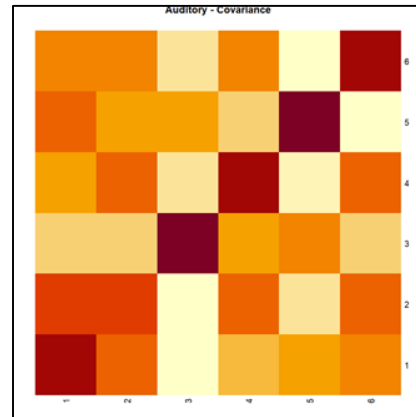
Symmetrized

PCA –
Eigenvalues

Auditory

Estimated
Regional
Marginal
Covariance

Visually



Thank you for your P300 Peak !

Thank you for your Attention !

Components in the
P300: Don't forget the
Novelty P3! –
SPECIAL ISSUE: FIFTY
YEARS OF
P300: WHERE ARE WE
NOW?

Received: 20 February 2018 | Revised: 27 February 2018 | Accepted: 28 February 2018
DOI: 10.1111/pspp.12371

SPECIAL ISSUE: FIFTY YEARS OF
P300: WHERE ARE WE NOW?

WILEY **PSYCHOPHYSIOLOGY**

Components in the P300: Don't forget the Novelty P3!

Robert J. Barry¹ | Genevieve Z. Steiner^{1,2} | Frances M. De Blasio¹ |
Jack S. Fogarty¹ | Diana Karamacoska¹ | Brett MacDonald¹

